

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Monday, 11 May 2026, 7:07 PM
State	Finished
Completed on	Monday, 11 May 2026, 7:46 PM
Time taken	38 mins 33 secs
Name	BHUVANA S 2025-BME

Question **1**

Correct

Marked out of
100Flag
question

Develop a **menu-driven program** to implement a **Stack using Array**. The program should support the following operations:

1. **Push (Insert an element into the stack)**
2. **Pop (Remove the top element from the stack)**
3. **Peek (Display the top element)**
4. **Display (Print all elements of the stack)**
5. **Exit**

Input Format:

- The first input represents the **choice of operation**.
- Based on the operation:
 - **Push** → Enter the element to be inserted.
- Continue operations until the user selects **Exit (5)**.

Stack Condition Rules:

- **Overflow:** When trying to push into a full stack.
- **Underflow:** When trying to pop from an empty stack.

Sample Input:

```
1
10
1
20
1
30
1
40
1
50
1
60
4
3
2
2
2
2
2
2
5
```

Sample Output:

```
Stack Overflow...!
50 40 30 20 10
50
50
40
30
20
10
Stack Underflow...!
```

For example:

Input	Result
1	Stack Overflow...!
10	50 40 30 20 10
1	50
20	50
1	40
30	30
1	20
40	10
1	Stack Underflow...!

50 40 30 20 10

50

50

40

30

20

10

Stack Underflow...!

For example:

Input	Result
1	Stack Overflow...!
10	50 40 30 20 10
1	50
20	50
1	40
30	30
1	20
40	10
1	Stack Underflow...!
50	
1	
60	
4	
3	
2	
2	
2	
2	
2	
5	

Answer: (penalty regime: 0 %)

Reset answer

```

1 #include <stdio.h>
2
3 #define MAX 5
4
5 int Stack[MAX], top = -1;
6
7 int IsFull();
8 int IsEmpty();
9 void Push(int ele);
10 void Pop();
11 void Top();
12 void Display();
13
14 int main()
15 {
16     int ch, e;
17     do
18     {
19         scanf("%d", &ch);
20         switch(ch)
21         {
22             case 1:

```

	Input	Expected	Got	
✓	1	Stack Overflow...!	Stack Overflow...!	✓
	10	50 40 30 20 10	50 40 30 20 10	
	1	50	50	
	20	50	50	
	1	40	40	
	30	30	30	
	1	20	20	
	40	10	10	
	1	Stack Underflow...!	Stack Underflow...!	
	50			
	1			
	60			
	4			
	3			
	2			
	2			
	2			
	2			
	2			
	5			

Passed all tests! ✓

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Friday, 15 May 2026, 5:40 PM
State	Finished
Completed on	Friday, 15 May 2026, 5:58 PM
Time taken	17 mins 49 secs
Name	BHUVANA S 2025-BME

Question **1**

Correct

Marked out of 1.00

Flag question

Develop a **menu-driven program** to implement a **Queue using Array**. The program should support the following operations:

1. **Enqueue (Insert an element into the queue)**
2. **Dequeue (Remove an element from the queue)**
3. **Display (Print all elements of the queue)**
4. **Exit**

Input Format:

- The first input represents the **choice of operation**.
- Based on the operation:
 - **Enqueue** → Enter the element to be inserted.
- Continue operations until the user selects **Exit (4)**.

Queue Condition Rules:

- **Overflow:** When trying to insert into a full queue.
- **Underflow:** When trying to delete from an empty queue.
- Follow **FIFO (First In First Out)** principle.

Sample Input

```
1
10
1
20
1
30
1
40
1
50
1
60
3
2
2
2
2
2
2
3
4
```

Sample Output:

```
Queue is Overflow...!
10 20 30 40 50
10
20
30
40
50
Queue is Underflow...!
Queue is Underflow...!
```

For example:

Input	Result
1	Queue is Overflow...!
10	10 20 30 40 50
1	10
20	20
1	30
30	40
1	50
40	Queue is Underflow...!
1	Queue is Underflow...!
50	
1	
60	

50
Queue is Underflow...!
Queue is Underflow...!

For example:

Input	Result
1	Queue is Overflow...!
10	10 20 30 40 50
1	10
20	20
1	30
30	40
1	50
40	Queue is Underflow...!
1	Queue is Underflow...!
50	
1	
60	
3	
2	
2	
2	
2	
2	
3	
4	

Answer: (penalty regime: 0 %)

Reset answer

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
#include <stdio.h>

#define MAX 5

int Queue[MAX], front = -1, rear = -1;

int IsFull();
int IsEmpty();
void Enqueue(int ele);
void Dequeue();
void Display();

int main()
{
    int ch, e;
    do
    {
        scanf("%d", &ch);
```

	Input	Expected	Got	
✓	1	Queue is Overflow...!	Queue is Overflow...!	✓
	10	10 20 30 40 50	10 20 30 40 50	
	1	10	10	
	20	20	20	
	1	30	30	
	30	40	40	
	1	50	50	
	40	Queue is Underflow...!	Queue is Underflow...!	
	1	Queue is Underflow...!	Queue is Underflow...!	
	50			
	1			
	60			
	3			
	2			
	2			
	2			
	2			
	2			
	3			
	4			

Passed all tests! ✓